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Challenges in Performance Analysis in Enterprise Architectures

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***Abstract*— This paper reviews the support for modeling KPI-related concepts in several enterprise modelling approaches and enterprise architecture frameworks. The scarcity of KPI modeling in EA approaches led us to open our scope of investigation to proposals in Business Process Management (BPM) area. Inside each approach, we first describe how these efforts propose to align goal-related concepts with KPI-related concepts (if the approach presents some kind of support for goal modeling). Further, the conceptual characterization for modeling KPI-related concepts in the context of EA models is also explored. Finally, we devote some considerations on how the KPI-related concepts are used to measure the properties of the EA elements and evaluate the achievement of goals. We conclude the paper proving challenges for the conceptual representation of a KPI modeling language that aims at measuring goal satisfaction the context of EA models.**

Keywords - Key Performance Indicators (KPIs), goal modeling, Enterprise Architecture, Business Process Management

I. INTRODUCTION

Mainly aiming at staying in business or seeking for higher profits, organizations today need support for fostering innovation and boosting production. To achieve these goals, the discipline of Enterprise Architecture (EA) [*1*] emerge as an enabler to capture and reason about the distinct dimensions or viewpoints [*1*] of the enterprise by means of *visual models*. Among these viewpoints, the domain of "motivation" has been recognized as an important element of enterprise architectures, also recognized by Zachman [*2*]. Goal modeling explicitly captures the goals of an enterprise, documenting an enterprise's strategy [*3*]. Capturing the enterprise goals is essential for business improvement once changes in a company's strategy have significant consequences within all organizational domains [*4*].

Although goal modeling in the context of EA efforts is arguably fundamental, little attention is devoted to explicitly represent goals and use goal-related concepts to increase the value of the enterprise modeling techniques [*5*]. This finding motivated us to investigate the current literature in goal modeling in [*6*], in order to understand the required characteristics of a goal modeling language that intends to represent goal-related concepts and relations of these concepts with the elements of the EA. In this effort, one of the main challenges comprehends the ability of expressing whether a given goal has been achieved or not. Moreover, it is also essential to represent the notion of partial goal satisfaction, for the cases in which a goal is not fully met. This quantification of goal satisfaction can be usually done by employing the concept of *Key Performance Indicator* (*KPI*) [*7*] [*8*]. KPIs have the characteristic of being able to evaluate the performance of organizational operations and subsequently use this information to measure the achievement of operational and strategic goals [*7*] [*9*].

In this paper, we review the support for modeling KPI-related concepts in several enterprise modelling approaches and EA frameworks. A survey in literature revealed that the KPI concept is largely explored in several topics of research, such as management sciences and process performance management. However, the support for modeling KPIs in the context of EA models is still incipient [*7*] [*10*]. With this finding in hands, we opened our scope to considerations in Business Process Management (BPM) and management sciences fields. Inside each approach, we first describe how these efforts propose to align goal-related concepts with KPI-related concepts (if the approach presents some kind of support for goal modeling). Further, the conceptual characterization for modeling KPI-related concepts in the context of EA models is also explored. Finally, we devote some considerations on how the KPI-related concepts are used to measure the properties of the EA elements and evaluate the achievement of goals.

The main contributions of this paper are twofold: (i) to provide an integrated overview of the support for modeling KPI-related concepts in existing enterprise modeling and BPM approaches. Furthermore, to explore how these concepts are used for the measurement of goals achievement and (ii) to identify gaps in these proposals for future research efforts concerning both the representation of KPI-related concepts as well as the usage of these concepts to measure the achievement of goals.

This paper is organized as follows: Section II discusses the topic of KPIs in some areas of research, providing criteria for the selection of approaches as well as criteria used to investigate inside each approach. Section III introduces the relevant aspects within our literature review, dividing these aspects into three sub-sections. The first sub-section introduces the integration of KPI-related concepts with goal-related concepts in modeling languages, whereas the second one deals with the KPI-related concepts, their characteristics and relations. Finally, third sub-section discusses the support for measurement of goal achievement through the measurement of KPI-related concepts. Section IV elaborates on challenges for a KPI language in the context of EA models that aims at measuring the achievement of organizational goals. Finally, Section V concludes the paper and outlines directions for further investigation.

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II. The Concept of KPI in Literature

Gartner defines the concept of Corporate Performance Management (CPM) as "processes, methodologies, metrics and technologies to measure, monitor and manage the performance of the business" [*11*]. This acronym is usually used to denote not only the long-term management of performance within the organization, but also includes process-oriented modes of action and approaches for the management of performance in business processes [*8*]. In this context, KPIs are defined as financial and non-financial metrics used to quantify the performance over time toward the meeting its strategic and operational goals [*8*] [*12*]. The topic of KPIs is explored in a number of areas in literature, such as management sciences, enterprise management and BPM. Through this section, we give a brief overview of the concept within these several areas of research.

Management Sciences. In management sciences, academic and practitioner circles concentrate both on identifying and classifying important performance indicators for companies (summarized in the format of *frameworks* or *indicators libraries*) [*7*] [*13*]. Examples of these frameworks are the best-known Balanced Scorecard [*14*], the Performance Prism [*15*], the Performance Measurement Questionnaire [*16*], the SMART System/Performance Pyramid [*17*] [*18*] and the EFQM Business Excellence Model [*19*]. These frameworks are usually composed by a number of indicators and a methodology that prescribes the required steps to derive indicators for a specific organization, by adapting the questions suggested in the library to some specific organizational context. While they provide a strong methodological support for suggesting, deriving and adapting indicators to a particular organizational scenario, the specification of indicators and the goals derived from these indicators are still done using natural language (and latter on, technically implemented) or using informal techniques (such as Strategy Maps [*20*] in Balanced Scorecard), without specifying important information concerning a goal-based measurement of indicators. A disadvantage of this approach is that the representation of the information that must be measured may present problems related with ambiguity, incompleteness and traceability once the indicators (and goals) are not represented through a (formal) modeling language. Regarding the computer support, we have the employment of Decision Support Systems (DSS), Executive information systems (EIS) or Business Intelligence (BI) projects that use Data Warehouse and OLAP tools [*8*], such those presented in [*21*] [*22*].

BPM. The focus of the management sciences approaches were originally target to collect only numerical, data-driven (mostly financial) indicators as means of obtaining a panorama of the overall costs and profitability of the organization [*7*]. This enterprise orientation has recently changed to accommodate a process-oriented performance management in the field of BPM [*8*] [*24*]. The focus became more concentrated in business processes, enabling the collection of a broader spectrum of performance-relevant data, covering both financial and non-financial measures [*8*].

This change in the orientation led practitioners to create a new acronym for KPIs, now denominated as PPIs (Process Performance Indicators) that represent process-oriented KPIs [*8*] [*25*]. PPIs are defined as quantifiable metrics that allow the evaluation of the efficiency and effectiveness of business processes. They can be measured directly on the basis of data generated within the process flow and are aimed at the process controlling and continuous optimization [*8*]. Activities related to the management of performance (the so-called KPI lifecycle [*8*]) have been also incorporated to the overall BPM lifecycle [*26*].

A remarkable trend in BPM refers to the tight representation of KPIs together with their correspondent technical implementation. This can be explained by the fact that business processes are usually accomplished by service-based applications in Service Oriented Architecture (SOA) approaches. As a consequence of that, processes are represented in terms of technical service composition and their measurement is realized through KPI monitoring of web-service compositions. This makes the KPI representation to be defined from an implementation perspective (at a very low level), too formal and/or close to the technical view, becoming thus hardly understandable to non-technical users [*8*] (an exception for that can be however found in [*8*]).

Regarding the KPIs measurement and monitoring in BPM, Complex-Event Processing (CEP) technologies [*27*] are used as means of calculating the KPIs values. They are useful for monitoring service composition process, once they use the events generated during service composition execution. CEP technologies can be roughly categorized into three groups according to the point in time in which KPIs are evaluated: (i) approaches for *post analysis* of process such as in process controlling [*28*] and process mining techniques [*29*], (ii) approaches for *real-time* monitoring of processes (using Business Activity Monitoring approaches (BAM) [*30*] [*31*] approaches) or (iii) approaches for *predictive analysis* as those used in process intelligence techniques [*32*] [*33*].

EA. Finally, it is possible to claim that the EA area is positioned in between management sciences and BPM fields, presenting characteristics from both types of approaches. On one hand, the area recognizes a holistic view of the organization (as management sciences) and does not expend a close focus into operational aspects. On other hand, the area is compromised with the explicit enterprise representation using *visual models* (also an assumption of BPM), but opposes to the informal treatment provided by the management sciences for the KPI representation.

Concerning the representation of KPIs in the area, although some EA approaches [*7*] [*25*] [*34*] [*35*] [*36*] acknowledges the existence of KPIs as means of quantifying the organizational performance, there is a considerable gap on modeling performance indicators and integrate them into modern enterprise modeling frameworks [*7*] [*10*]. As consequences of that, the automated support for KPIs measurement is seriously impaired as well as the measurement of goal achievement.

After this brief overview in the literature for KPI representation and methodology to understand the current state of art, we are in position to draw general conclusions about the literature in the topic. They can be explained as follows:

- **Approaches in Management Sciences.** The approaches have been initially motivated by the concern of providing frameworks (set of indicators) in the context of management of organizations. However, the limited support for KPI representation (informal) led us to exclude these approaches for the time being. However, we believe that future efforts in methodologies for KPI elicitation and refinement can fruitfully employ these frameworks;
- **Approaches in BPM.** As previously argued, the approaches are more oriented towards representing operational and short-term aspects of process measurement [37]. Even with this shortcoming, most of approaches that represent KPIs in the format of *modeling languages* pertain to the BPM area, reason why we have included these approaches in our survey;
- **Approaches in EA.** Since we focus our understanding in how KPIs-related concepts are represented within EA models, this is the most important area to ground our investigations. In a previous effort [5] we investigated the support for modeling goal-related concepts in the context of enterprise modeling languages and frameworks. These approaches have also been included here.

Based on these conclusions, we have selected some approaches in literature that support our research goal. Table I gives an overview of the approaches used in this survey.

Table I Approaches used in our literature survey

Approaches
Momm et al. Approach [38] [39]
COBRA/SENTINEL Approach [23] [40] [41]
IBOM Approach [32]
Mayerl et al. [42]
PPINOT Approach [8]
Popova et al. Approach [7] [43] [44]
Wetzstein et al. Approach [31] [45]
MEMO Approach [34]
ARIS Approach [8] [46] [47]
URN Approach [9]
BIM Approach [8] [48]
MMC Approach [8] [49]
Friedenstab et al. Approach [8] [50]
Birgit et al. Approach [51]
Zachman Framework [2]
ISO RM-ODP Enterprise Language [5] [52]
DoDAF Framework [5] [53] [54]
MODAF Framework [5] [55]
BMM [5] [56]
ArchiMate [5] [35] [57]
Matthes et al. Approach [10]
Federated Enterprise Architecture (FEA) [58]
Australasian Government Architecture [59]

This next section summarizes the relevant aspects found when one engages in modeling indicators as means of measuring the achievement of goal in EAs. The aspects are divided into three main sub-sections and each of them examines the BPM and EA approaches with respect to the following criteria[1]:

- The support to associate the KPI-related concepts with goal-related concepts in the languages;
- The support for representing KPI-related concepts in the proposals (through the use of *modeling languages*);
- Once goals and KPIs are represented in a suitable manner, we extend our considerations to assess how the concept of KPI can be used to measure the achievement of goals.

The analysis of each approach has been guided by idea of including all the attributes used to build our conceptualization in terms of these three aforementioned aspects. As a consequence of that, given the fact that approaches usually make such characterization using a common core of attributes (with slight variations), some attributes are cited by several sources, while others have a reduced number of them. We believe that even slight variations contribute to build a more solid conceptual foundation to the phenomena under consideration. Further, we also present an illustrative example to motivate the problem from a real-world perspective and to guide the reader in the terminology of the domain.

III. Current Support for KPI Modeling for Measuring Goals in EA

We start our considerations by introducing a running example of an insurance company that provides insurance services to its customers. In this company, the Perform medical expert examination to evaluate refund benefit process has the main aim of evaluating if some insured person has the right of being refunded due to expenses with medical treatment. For problems with performance, the senior management decided that the efficiency of the process should be increased and thus created the Increase the efficiency of process of refund benefits goal.

We pass to the literature description, demonstrating how the approaches meet or fail in supporting the three

[1] In [8], the reader is referred to find an extensive literature review in terms of KPI representation within the BPM discipline. We incorporated some of the insights of the approach to understand the conceptual definition of KPIs, but also extended this evaluation to the EA field.

aforementioned aspects and using this running example when necessary for illustration purposes.

A. ***Goal modeling.*** *Integration of KPI-related concepts with goal-related concepts in modeling languages*

The integration of the KPI languages with languages for goal modeling is the first requisite for the numeric quantification of goal achievement in EA approaches.

Exemplifying this integration by means of our running example, the Increase efficiency of the refund benefit process goal should be represented together with the refund benefit process. Furthermore, this goal must be measured accordingly through the use of KPIs. Usually, methodologies for KPI discovery [*28*] state that one or more KPIs can be derived from the same goal, depending on the way one has to measure the achievement of this goal. For instance, in our running example, efficiency can be conceived in two different ways: it can characterize performance in terms of *quantity* (thus we create the Number of evaluations of refund benefit KPI) or in terms of *time* (Average time spent in the refund benefit process KPI). In this case, whether we decide to represent one or more KPIs, the modeling language should support us in our intent.

Concerning the support for this representation in the current literature, unfortunately, the support for modeling KPI-related concepts and integrate these concepts with concepts in goal modeling languages is not extensive (a fact also found in [*60*]). We have basically two groups of proposals: (i) proposals that do not recognize the existence of goal-related concepts integrated with constructs of the KPI modeling language [*2*] [*32*] [*38*] [*42*] [*45*] [*49*] [*50*] [*52*] [*55*] [*56*] [*58*] [*59*] and (ii) approaches that recognize a very simple 1:1 relation between goals and KPIs (1 goal can be associated to 1 KPI) [*7*] [*8*] [*9*] [*10*] [*34*] [*41*] [*46*] [*48*] [*51*] [*54*] [*57*]. Proposals that allow the representation of more complex relations between goals and KPIs should support the modeling of examples as depicted in the methodological steps mentioned in this section.

B. ***Characteristics for the language of KPI modeling.*** *Concepts and relations for KPI-related concepts representation in the context of EA models*

Once the goals are represented, the next step comprises in representing their associated KPIs. In literature, a conceptual definition of KPIs can be made in terms of the definition of the concept of KPI, their characteristics (or attributes) and the relations between the KPI concept. Therefore, our description of literature is divided in terms of these three elements:

Concept of Key Performance Indicator (KPI). KPIs are defined as financial and non-financial metrics used to quantify the performance over time toward the meeting of its strategic and operational goals [*8*] [*12*].

KPI attributes. Many proposals acknowledge the need of representing KPI's attributes through the SMART criteria [*61*]. SMART is a mnemonic used to set objectives and KPIs/PPIs [*8*] and stands for the following words: **Specific** (it must be clear what the KPI specifically describes), **Measurable** (it has to offer the possibility of measuring the current value and compare with the target one), **Achievable**, **Relevant** (it must be aligned with organizational strategy) and **Time-bounded** (it only has a meaning if it is known the period of time it is measured). In the remainder of this section, we enumerate the attributes that we found relevant in literature to characterize KPIs, highlighting the attributes that meet the SMART criteria. For the sake of exemplification, we choose the Number of evaluations of refund benefit KPI from the running example to properly explain the terminology.

Basically speaking, we found that a KPI can be characterized by a **name** and **description** [*8*] [*32*] [*34*] [*58*] [*44*], **scale** [*38*] [*53*] [*55*] [*58*] [*44*], **measure** [*8*] [*10*] [*23*] [*32*] [*34*] [*38*] [*55*] [*58*] [*44*], **current value** [*10*] [*34*] [*38*] [*53*] [*55*] [*58*], **target value** [*8*] [*9*] [*10*] [*32*] [*38*] [*45*] [*53*] [*55*] [*58*], **threshold** [*9*] [*38*] [*10*] [*45*] [*58*] [*44*] [*23*] [*62*], **source** [*44*] [*9*] [*34*], **frequency** [*10*] [*58*] [*62*], **responsible** [*8*] [*58*], **informed** [*8*] and **owner** [*7*] [*10*] attributes. The specification of the following attributes allows the proposals to meet the SMART criteria, respectively: Specific and Measurable (**measure**), Achievable (**target value**), Relevant (**goal**) and Time-bounded (**scope**).

A **measure** attribute defines how the KPI values are calculated by measuring the attributes of the EA elements. For example, since we decided to choose the Number of evaluations of refund benefit KPI to be implemented in the refund benefit process, a measure for this KPI is a particular way of measuring the attributes of the refund benefit process (EA element), using data generated during process execution. From a methodological point of view, from one specific KPI, several different measures can be created, i.e., the number of evaluation measure may correspond to the different measures within the refund benefit process. For example, when the process executes, it issues whether a given request of refund is accepted or not. Therefore, the number of evaluation measure may correspond to all requests submitted in a period interval of 1 month. Alternatively, it may correspond to only those requests that have been made by the customers with less than 60 years old in an interval of time of 2 months. We decided to choose the second measure to implement in the process (number of evaluation measure made by the customers with less than 60 years old in an interval of time of 2 months).

Once we have implemented this measure, a **current value** for the KPI may be calculated. The current value represents the value of the indicator that is measured in the current time and calculated on the basis of runtime information. This current value must be compared with the **target value** that stipulates an ideal value to be reached. In our example, after implement the measurement, we discovered that the current value is number of evaluations = 80 evaluations. With respect to the target value, while our goal states that the performance should be increased, it does not specify a quantitative value for that. Based on business needs, we decided that, in order to increase the performance, the target value should be set as number of evaluation measure > 100 evaluations. A useful distinction that we found in literature [*8*] corresponds to the concepts of Simple, Composed and Custom targets. A **Simple Target** is used to specify the lower bound and/or upper bound that make up the range within which the KPI value should be, i.e. the KPI admissible value range. A **Composed Target** enables the definition of several target values or ranges, for those cases where the value of the KPI is discrete. For example, if we model the "evaluation" as a data object in our example, only those evaluations whose the attribute age of customer < 60 years old should be considered for the calculation of the measure. In this case, we have a composed target value, where number of evaluations > 100 evaluations and age of customer < 60 years old (where age of customer is an attribute of a data object represented in refund benefit process). Finally, a **Custom Target** offers the possibility to set a restriction that the KPI value must fulfill. We also have to specify the KPI **threshold** that represents the deviation in relation to the target value that is acceptable for the current value to reach. In our case, it is admissible to consider number of evaluations > 90 evaluations and age of customer <= 60 years old.

Other relevant attributes to be represented are the source, frequency, responsible, informed and owner attributes. The **source** attribute comprehends the internal or external sources used to extract the performance indicators, such as company policies, mission statements, business plan, job descriptions, laws, domain knowledge, etc. Alternatively, the sources can also represent the systems where the data comes from for the calculation of the KPI [*9*] [*34*]. The **frequency** represents the frequency of KPI measurement during a specific interval of time. The last three attributes (responsible, informed and owner) link the KPI to the organizational structure. The **responsible** attribute represents the role, department or organization that is in charge of measuring the KPI, whereas the **informed** represents the agent, roles, departments or organizations that are somehow interested in the KPI. Finally, the **owner** can be defined as a stakeholder in the enterprise that is responsible for the achievement of the defined KPIs.

KPI relations. This dimension represents the relations that KPIs may present among themselves:

- **Correlation relation** [*43*]: when KPIs are correlated, a change in one KPI implies a change in the same direction of the other KPI (i.e., in a *positive correlation*, if one KPI increases the other KPI also increases, in a *negative correlation*, if one KPI decreases, also the other one);
- **Causality relation** [*43*] [*58*]: A change in one KPI causes a change in another one;
- **Conflict relation (correlation, causality, derivation in a negative way)** [*43*]: indicates that two requirements cannot be satisfied at the same time, what implies that the KPIs are related by either correlation or causality relations in a negative way;
- **Independence relation** [*43*]: when no significant relationship exists between KPIs;
- **Trade-off relation** [*43*]: when a measurement of one KPI is easily replaced by the measurement of another KPI, if is necessary;
- **Is_costlier_than relation** [*43*]: when the information that the measurement of one KPI conveys is the same that the information of another one, but the measurement of the first one is more expensive to obtain, then it is said that the first KPI *is costlier than* the second one;
- **Customized relation** [*34*]: enables the qualification of further customized relations.

*C. **KPI Measurement.** Relationship between the KPI-related concepts with the EA elements*

Once we represented the KPIs and their correspondent relations, the next step comprehends the measurement of the KPI values during runtime in order to evaluate if the goals are being achieved or not. To perform the measurement of KPIs, the proposals use the concept of **measure** presented in the definition of the KPI concept. The measures represent a particular way of colleting information about the properties of the elements of the EA to obtain the value of a given KPI. When formulating measures for KPIs the proposals distinguish into the following dimensions: the number of instances, the type of measure, the type of scope, the type of monitoring objects and monitoring points. Each one of these dimensions is explained in the remainder of this section.

The **number of instances** dimension refers to the number of instances that are necessary to calculate the KPI's value. A **base measure** (or instance measure) is the measure obtained executing a certain measurement method over a single process instance, whereas an **aggregated measure** [*8*] [*38*] [*32*] [*45*] [*42*] [*48*] [*23*] [*43*] [*47*] [*50*] [*49*] uses a set of process instances to calculate the KPI's values (aggregating single process instances). The aggregation is usually performed by selecting all the instances in a given interval of time (called as *analysis period* in a temporal aggregation), but also by selecting all the instances for a specific employee/unit/company (organizational aggregation) [*43*] and afterwards applying a certain aggregation function like max, min, average, sum on this set of measures to obtain one single value [*45*] [*63*]. In our example, the number of evaluations measure does not specify the number of instances used to calculate the KPI value. Since we intend to evaluate the efficiency of the process in a given period of time, we stipulate that the number of evaluation should be collected for a given set of instances in a period of 2 months (analysis period) and the number of evaluations is the average number of evaluations obtained in the time interval of 2 months. Therefore the current value number of evaluations = 80 evaluations is actually an average number of evaluations (for the customers with less than 60 years old) between January and February. In this dimension, it is also possible to specify **derived measure** [*8*] [*32*] [*45*] [*42*] [*50*] [*49*] [*48*] [*23*] [*47*] [*43*] that corresponds to measures defined on the basis of other measures (either base or aggregated measures). Their value is calculated by performing a mathematical function to combine two or more measures.

The **type of measure** dimension concerns the type of the property that needs to be measured. A **time measure** [*8*] [*38*] [*32*] [*45*] [*42*] [*50*] [*49*] [*48*] [*23*] [*43*] [*47*] measures the duration of time between two time instants (these two time instants can correspond to the change to a certain state of a process element like an activity or data object, or with the triggering of a certain event). When it is necessary to take measures between elements located within a loop, two ways of measuring time can be considered, a **linear measure** and **cyclic measure** [*8*]. The **linear measure** is defined as a measure that takes into account the first occurrence of a given time instant condition and the last occurrence of the other time instant condition. The **cyclic measure** takes into account pairs of time instant conditions that are located in a loop. In the end, the cyclic measures compute the KPI measure by aggregating the time values of all the iterations. A **count measure** [*8*] [*38*] [*32*] [*45*] [*42*] [*50*] [*49*] [*48*] [*23*] [*47*] measures the number of times something happens, where "something" may be represented by a change to a certain state of a process element or the triggering of a certain event, whereas a **condition measure** [*8*] [*45*] [*48*] measures the fulfillment of certain condition in running or finished process instance. Finally, a **data measure** [*8*] measures the value of a certain part of a data object. It is possible to specify a condition that a given data object must fulfill when the measure is performed (for example, the data objects has to be in certain state). A **human resource measure** [*7*] describes the performance of a role or agent. To better exemplify these distinctions, in our number of evaluations measure, we have to specify that the number of evaluations correspond to count all the times that a start event is triggered (to count all the evaluations using a count measure). In addition to that, we use a data measure on the evaluation data object (data measure) by specifying a condition that the age of customer attribute is less than 60 (condition measure age of customer < 60).

The **type of scope** [*8*] dimension has been implicitly discussed along this text and represents some sort of filter that selects a set of process instances used for the computation of KPI's values. The instances can be selected by the following scopes: a **number of instances** [*8*] [*23*] [*45*] [*47*] [*50*] that selects just a number of instances for the calculation of a given KPI, by a **temporal scope** [*8*] [*43*] [*23*] [*50*] [*55*] that selects the instances that are in a given interval of time and a **process state scope** [*8*] [*43*] [*23*] [*47*] [*50*] that selects the instances that are in a given process state. In our example, we have opted for selecting the instances in a given temporal scope (analysis period) of two months.

Once defined the aforementioned measure attributes, the definition of the **monitoring objects** [*38*] [*32*] [*34*] [*45*] [*53*] [*64*] is also essential to enable the measurement of the EA elements. The monitoring objects represent the association between the KPI language and the other viewpoints of the enterprise. In the scope of goal modeling approaches, monitoring objects are usually used interchangeably as "architectural elements" or "EA elements" terms. The monitored objects represent the architectural elements whose properties are measured. They are the elements where the monitoring operates on to have the KPI's values determined. We found that the following objects are amenable of measurement:

- Processes (and activities) [*8*] [*34*] [*38*] [*42*] [*45*] [*32*] [*43*] [*23*] [*47*]
- Gateways [*38*]
- Data objects (attributes of the data object, data object state, data object state change) [*8*] [*34*] [*49*]
- Human resources [*43*] [*47*]
- Software and hardware [*55*]

To perform the actual measurement of the monitoring objects, the proposals use the concept of **monitoring points** [*65*] [*38*] [*47*]. Monitoring points define points in the process where data comes from for the KPIs computation [*47*] and can be attached to any monitoring object on which it is intended to gather information. In the case of our running example, we have two options where we may attach the monitoring points. First, we can count how many evaluation data objects exist. This corresponds to attach a monitoring point on the evaluation data object. Alternatively, we may attach the monitoring point on the start of the refund benefit process, in order to know how many times the process has started (how many process instances have been created) and thus, obtain an indirect measure. The second action is to decide how to obtain the measures that are necessary to calculate the KPIs using the available information systems.

IV. Challenges

Along the trajectory of description of literature, we have identified gaps of research that are not addressed (or partially addressed) by the aforementioned approaches. In this section, we enumerate these gaps in the format of research questions and link them with the sections used for the literature description, when necessary.

A. How many levels of abstraction for goals must be represented?

Potentially, the definition of the goals may be related to a number of aspects within the enterprise environment. Usually, goal statements range from high-level concerns in an enterprise (expressing the vision and mission of the organization) to declarations of the values that must be achieved by process execution on behalf of stakeholders. Therefore, the **level of abstraction** dimension aims at classifying goals with respect to the aspects they can potentially address within the organizational environment. To exemplify this category, in section III.A, we presented a goal (Increase the efficiency of process of refund benefits goal) which was particularly associated to one specific process. Intuitively speaking, higher-level goals are necessary, such as goals related to several processes or other aspects such as organizational mission and vision (also treated as fundamental goals in other proposals [*66*]).

In current literature for goal modeling [*67*], few proposals recognize the need of such distinction, although some exceptions can be found [*41*] [*56*] [*66*]. A problem with the approaches that acknowledge such distinction is related to the lack of precise criteria for allocating goal statements into the categories suggested by the proposals. For example, considering the insertion of an additional goal in our running example, such as Increase efficiency of organizational services goal, the challenge is to determine whether this goal fits into a *process goal category* (as one of the levels of abstraction), or should be classified in the *fundamental goal category* (other level of abstraction).

A potential solution to this issue may involve the actual need of creating the levels of abstraction. It is largely recognized that in EA practical efforts, goals are used as requirements for generating a target architecture. Thus, the challenge is to discover how many levels of abstraction are useful for guiding this decision-making process from a practical point of view. For instance, if we need to create the level of abstraction categories to enable a business process management initiative to better manage its business process based on goals, we could associate each goal level of abstraction with each level of our process landscape [*65*]. Therefore, we could classify the Increase the efficiency of process of refund benefits goal as a *process goal* (process goals are the lowest level of refinement of a goal structure in our example) and the Increase efficiency of organizational services goal as a *fundamental goal* (upmost goal category in the structure). The intermediate levels of abstraction for goals are determined by associating them with particular levels of abstraction within the process landscape.

B. Once we have several levels of abstraction for goals, which levels of abstraction should be associated with KPIs?

In section III.A, we have argued that goals are usually associated with KPIs in a 1:1 relation in the current literature (1 goal is associated with one indicator). With several levels of abstraction, a question that certainly arises is to which levels of abstraction the indicators should be associated.

Unfortunately, if we try to find a response of this question in the surveyed literature, we realize that although the proposals link goals to indicators (in a 1:1 relation), most of them do not recognize the need of allocating goals in different levels of abstraction. On the contrary, those proposals that make such distinction, they do not provide criteria for goal satisfaction based on the evaluation of KPI's values, such as [*41*] [*56*] [*66*]. Two alternatives may be envisioned for this issue, but further research is still required to reach a final solution. In the first solution, the goal structure is segmented into several levels of abstraction and for each goal in the structure one KPI is associated. Alternatively, only *process goals* (leaf goals) are associated with KPIs and the evaluation of higher-level goals is performed through the evaluation of *process goals* and subsequent propagation of goal satisfaction to the higher levels of the structure.

C. How to relate the goal domain with the other EA domains?

To consider the measurement of goal achievement, the first concern is to identify the linkage of the goal domain with the elements of the EA. In section III.C, we mentioned an association of KPIs with the elements of EA through the concept of monitoring objects. A linkage between the goal domain with the other EA viewpoints should be further characterized, once many approaches refrain from addressing such issue, as in [*57*] [*67*]. Those proposals that recognize such integration adopt a restrictive solution, mainly relying on capturing the relationship between goal models and business process models [*4*] [*3*] [*68*], failing in addressing the other viewpoints of the EA.

Therefore, the monitoring objects used in this survey can be a good starting point to relate the goal domain with other enterprise domains. Moreover, further investigation could reveal other important monitoring objects to be included in such integration. In particular, we have already worked towards achieving this second goal (identification of other monitoring objects to relate the goal domain and other viewpoints of EA and their correspondent relations [*69*]). However, no contribution has been provided to properly represent these findings in EA models.

D. How to measure attributes of other monitoring objects?

Our survey revealed the concept of measure and how this can be used to perform measurements in the process' properties (section III.B). This concept has been proved to be very useful for process measurement, but could be also extended to measure the properties of the other elements (monitoring objects) of the EA, such as roles and software.

E. How to evaluate goal achievement using the KPI modeling language?

Goals need to be evaluated with respect to their achievement. This means that they also have a *goal lifecycle*, in which goals are created and represented accordingly, operationalized through a linkage with the EA elements and evaluated with respect to their achievement. We have noticed that, although the approaches associate goals and KPIs in the representation phase, the criteria for goal satisfaction based on KPIs is roughly explored in literature (the only exception is found in [*7*]).

To specify this criterion, the **evaluation type** [*7*] concept can be used. The evaluation type specifies how the satisfaction of the goal must be checked for a given interval of time. For example, while some goals have to be achieved, others must be denied instead. The evaluation types that we found relevant are: achieve/cease, maintain/avoid, increase/decrease [*6*].

Observe that the evaluation type assumes a given interval of time in which the satisfaction of the goal must be checked. This characteristic fits with the scope dimension of measures that selects only a given set of instances to compute the KPI's value. In other words, one could select only a portion of instances (that pertains to a given scope) to check the fulfillment of some goal.

V. Conclusions and Future Work

Methods for enterprise modeling address strategic concerns with the purpose of facilitating the decision-making process that determines how enterprise's objectives are operationalized. In this paper, we have surveyed how current enterprise modeling/frameworks and BPM approaches address the support for modeling KPI in order to support the measurement of these enterprise's objectives. During the realization of our survey, we have identified three main relevant aspects when one engages in using KPIs to measure the achievement of goals in EAs. These aspects have been structured in three different sections and the current approaches have been positioned in terms of meeting or failing in addressing such aspects. We do not claim to be exhaustive in covering all the aspects and approaches in our literature description, although we believe that a somehow representative scenario has been devised after the conclusion of this effort.

It is important to emphasize some aspects of our literature review. The first aspect concerns the coverage of approaches included in this survey. We included approaches that focus on modeling languages for KPI-related concepts representation (and the support for integrating the KPI-related concepts with goal-related concepts). Although languages for representing goal-related concepts are considered equally important to gain a solid understanding of the area, further analysis about the topic is out of the scope of this paper. More insights can be found in some of our previous efforts [*5*] [*6*] [*69*].

Among the surveyed approaches, some trends are noticeable. First, the terminology may present some variations within the area and other terms different from those used in this paper may also be found, such as "metrics", "measure type", "performance", "performance measurement", among others. However, this apparent lack of uniformity was a very preliminary insight and a deeper analysis revealed a common core of concepts (with some terminology variations) as those presented in this paper.

As a general conclusion we can observe that current approaches for enterprise modelling offer rudimentary support for KPI-related modelling as well as for the incorporation of these KPI-related aspects in the other viewpoints, a conclusion also corroborated by [*10*]. This can be evidenced by the fact that some approaches do not provide support for capturing a number of aspects mentioned in the literature description. Just as a brief example of that, we may cite frameworks like DoDAF (SV-7 Systems Measures Matrix) [*53*], MODAF (SV-7 Resource Performance Parameters Matrix) [*55*] and Australian Government Architecture [*59*] (Performance Reference Models). We may notice that they have been barely cited with respect to the support provided to the aspects raised in the literature review.

We can conclude that the problem of capturing KPI-related in accordance with strategic concerns is an issue commonly neglected in the current languages and frameworks, motivating us to set up a research agenda into two promising veins.

The first direction concerns the creation of modelling languages to capture the relevant concepts acquired through our literature study. In that respect, two alternatives are envisioned. First, a KPI modelling language with the aspects found in literature would represent a meaningful contribution to the EA practice. A second and more meaningful contribution would encompass both aspects from the literature review as well as the aspects enumerated as challenges. However, to work towards promoting this second goal, one would have to deal not only with the representational aspects of this modelling language, but also with other aspects of the knowledge represented in the model. Briefly discussing that, since the KPI representation is mainly aimed at capturing performance aspects within the EA, to deal with the concept of Service Level Agreements (SLAs) in the context of EA models seems a reasonable approach. In the BPM area, the KPI measurement is achieved by measuring attributes of processes that are realized as services compositions. Therefore, SLAs actually express agreements between *service requester* and *service providers* in the context of web-service provisioning. If we take as inspiration this idea and consider that KPIs measure the SLAs at the process level, then it is natural to argue that the process goals actually represent the business requirements that express the SLAs. This idea may be extended to represent SLAs of other EA viewpoints, by expressing goals and KPIs of these viewpoints.

The second direction for future work concerns the creation of methodologies for goal/KPI modelling and their integration into enterprise architecture and enterprise modelling approaches. Into that respect, the literature description of this paper can provide some hints. Our literature description is intrinsically performed in a *top-down* fashion, i.e., we start by modeling the relevant goals, deriving (modeling) their correspondent KPIs and associating these KPIs with the elements of the EA. This description may indicate a methodology for goals and KPI *representation*. A methodology for goal and KPI *measurement* is also relevant for future practical EA efforts, but has not been mentioned in the scope of this paper. In that respect, more research that aims to bridge this gap can be found at [*70*].

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